

RAKSHAKAI

Predictive Personnel Stress & Welfare Monitoring System

Smart India Hackathon 2026	Problem Statement: SIH26186
Team: ISM Troopers	Team ID: 188472
Category: Software	Theme: HealthTech / BioTech / MedTech

PROJECT REPORT

RakshakAI is presented as a privacy-first personnel welfare platform that combines daily wellness check-ins, stress and fatigue indicators, longitudinal trends, role-specific workflows, welfare intervention, clinical escalation, and operational readiness views in a single interface.

Primary project website: spal01-jpg.github.io/rakshak-ai/

Document basis: supplied SIH26186 problem statement plus the deployed RakshakAI web prototype.

1. Executive Summary

RakshakAI addresses the need for earlier identification of personnel stress, fatigue, burnout and welfare concerns. The supplied SIH concept describes a Digital Welfare Twin that combines operational data, wellness assessments and behavioral indicators, together with an AI Risk Assessment Engine and a Welfare Intelligence Platform. The deployed website demonstrates these ideas through a role-based web interface and simulated operational workflows.

The prototype homepage presents four primary goals: monitor well-being, predict risks, enable early intervention, and build a stronger force. It also states that monitoring is intended to be non-punitive and that risk prediction uses AI-driven early warning and behavioral analytics.

The interface supports four operational roles: Personnel, Medical Officer, Welfare Officer and Commander. Each role is associated with a distinct workflow, helping separate personal wellness information from command-level readiness information.

2. Problem Statement & Motivation

The project is based on SIH problem statement SIH26186, titled "AI-Based Predictive Personnel Stress and Welfare Monitoring System for Uniformed Forces." The SIH material identifies the core need as proactive welfare monitoring using deployment history, leave utilization, duty schedules, transfer frequency and workload patterns, alongside wellness assessments and optional physiological/activity signals.

The proposed approach is preventive rather than solely reactive: identify emerging patterns, provide confidential support pathways, enable welfare officers to validate cases, and give commanders aggregated readiness and workload insights without exposing unnecessary sensitive personal information.

3. Project Objectives

Objective	Project interpretation
Early risk identification	Detect patterns associated with stress, fatigue, burnout and welfare concerns before escalation.
Confidential self-monitoring	Provide personnel with daily wellness check-ins and a non-punitive interface.
AI-assisted decision support	Use risk analytics, explainability and rules to prioritize cases and recommend next actions.
Structured intervention	Connect personnel, welfare officers, medical officers and commanders through role-specific workflows.
Readiness visibility	Provide unit-level trends, workload distribution and readiness indicators while limiting sensitive details.
Privacy by design	Use consent, role-based access, anonymization, encryption, audit logs and separation from service records.

4. System Overview

The system can be understood as five connected layers: input and data capture, personnel wellness profiling, AI/rule-based risk assessment, intervention workflows, and role-specific dashboards. The SIH material explicitly names HRMS data, a wellness application and optional wearables as input sources, followed by a Digital Welfare Twin, AI Risk Assessment Engine and Welfare Intelligence Platform.

Layer	Main elements	Purpose
1. Input	HRMS; wellness check-ins; optional wearables	Collect operational, self-reported and optional physiological/activity signals.
2. Profile	Digital Welfare Twin	Maintain a continuously updated wellness profile.
3. Intelligence	AI models + rule-based analytics	Estimate risk and explain contributing factors.
4. Intervention	Welfare Officer + Medical Officer + Commander workflows	Validate severity, coordinate support and authorize/route actions.
5. Insight	Personnel, Welfare, Medical and Commander interfaces	Deliver the appropriate level of information to each role.

5. Key Features Demonstrated by the Website

Feature	Demonstrated capability
Personnel wellness check-in	Mood selection, sleep-hours input, stress indication, operational-strain selections, optional reflection and confidential submission.
Risk indicators	Stress, burnout, fatigue and welfare risk indicators plus a multi-dimensional resilience radar and longitudinal trends.
AI diagnostic attribution	The interface shows contributing factors such as leave deprivation, night-duty load, family pressure and physiological drift as example explanations.
Commander dashboard	Personnel counts, critical-stress counts, treatment counts, pending approvals, force-readiness indicator and operational filters are shown.
Medical Officer console	Clinical queue, referral categories, treatment status and clinical triage isolation are represented.
Welfare Officer console	Stress validation, advisory dispatch, treatment logs and a threshold-based routing workflow are represented.
Resilience center	The site includes tactical breathing, post-patrol muscle relaxation and sleep-hygiene content as recovery-oriented UI modules.
Mitra AI interface	A confidential Hindi/English conversational-support interface is presented as a prototype feature.
Offline / installable editions	The site advertises a Windows desktop package, PWA installation and a standalone offline HTML edition.

6. End-to-End Workflow

Step	Process	Result
1	Personnel enters a daily wellness check-in and relevant operational strain information.	Current wellness snapshot
2	Operational, wellness and optional sensor/activity inputs are combined.	Unified feature set
3	The Digital Welfare Twin updates the longitudinal personnel profile.	Continuously updated profile
4	AI and rule-based analytics assess stress, fatigue, burnout and welfare patterns.	Risk indicators + attribution
5	Risk thresholds and welfare rules classify cases for appropriate routing.	Priority / protocol path
6	Welfare Officer validates severity and either routes a treatment request or submits a commander application, according to the prototype workflow.	Intervention request
7	Medical Officer reviews referred cases and records clinical treatment status.	Clinical care workflow
8	Commander receives appropriate welfare/readiness actions and unit-level insights.	Operational decision support
9	Intervention outcomes and follow-up information feed the continuing welfare-monitoring loop.	Continuous monitoring

7. Role-Based Workflow

Role	Primary responsibilities shown in the prototype
Personnel	Daily check-in, stress assessment, wellness trends, confidential support request and recovery tools.
Welfare Officer	Review live stress ratings, validate severity, send advisories, manage welfare applications and coordinate support.
Medical Officer	Review referred personnel, determine/record clinical treatment and synchronize treatment status with welfare.
Commander	View force-level readiness and operational indicators and act on welfare recommendations/approvals.

8. Risk Assessment & Explainability

The website presents multiple risk dimensions—stress, burnout, fatigue and welfare—and displays longitudinal trends and a resilience radar. It also includes an AI Diagnostic Risk Attribution section that explains an example risk state using several contributing factors.

The SIH technical approach specifies explainable AI and rule-based analytics, while the technology-stack proposal lists XGBoost, LSTM and NLP/BERT as candidate model families. These should be treated as the proposed analytical stack from the submission; the supplied sources do not establish that every listed model is already trained or deployed in production.

9. Technical Architecture

Layer	Proposed / referenced technology	Use
Frontend	React / React Native	Mobile and web UI.
Backend	Node.js, Express, FastAPI	API/service layer.
Database	MongoDB, PostgreSQL	Application and operational data persistence.
ML & Analytics	XGBoost, LSTM, BERT/NLP	Risk prediction, temporal patterns and language analytics.
Risk layer	Risk scoring + alert generation	Prioritization and routing.
Dashboard	Welfare / Commander UI	Decision-support views.

Prototype and deployment characteristics

The accessible website is a web-based prototype and exposes a number of deployment-oriented features. Its download section describes a Windows desktop package with offline capability, a PWA installation path, and a standalone offline HTML edition. The site states that the offline edition is intended to run without Python, Node or server setup and includes an embedded personnel dataset and offline heuristic prediction.

These deployment descriptions should be understood as prototype/application packaging claims visible on the website. They are not, by themselves, evidence of a production security certification or field deployment.

10. Privacy, Security & Trust Model

Control	Purpose
Role-based access	Show only the information required by each operational role.
Consent management	Give personnel an explicit welfare-monitoring participation choice.
Non-punitive framing	Position self-reported wellness information as welfare-support data rather than disciplinary data.
Air-gapped welfare concept	The proposal describes separation of stress metrics from ACRs and disciplinary files.
Anonymization & privacy-preserving analytics	Reduce exposure of sensitive personal information in analytical outputs.
Encryption & audit logs	Protect information in transit/at rest and maintain accountability.

The website explicitly presents a voluntary digital-welfare consent section and states that self-reported mood, sleep and emotional-health indicators are intended for welfare support and preventive care.

11. Feasibility & Mitigation

Challenge identified in SIH material	Proposed strategy
Trust deficit and reporting stigma	Cryptographic Privacy Shield and explicit welfare-monitoring consent.
Biometric noise / environmental drift	Cross-analysis of physiological signals with resilience history, leave deprivation and night-shift frequency.
Disconnected hierarchy	Welfare Officer Air-Gapped Bridge.
Command approval delays	75% Rapid Triage Protocol described in the proposal.
Immediate field decompression	In-situ resilience tools such as breathing and post-patrol relaxation flows.

12. Expected Impact

Area	Expected benefit stated in the SIH material
Proactive welfare	Earlier identification and intervention for personnel needing support.
Data-driven planning	Evidence-based resource allocation, training and duty scheduling.
Well-being & resilience	Improved mental well-being and workforce resilience.
Workload management	Better workload distribution and personnel management.
Operational readiness	A healthier, more resilient workforce with fewer fatigue-related incidents.
Scalability	Adaptable concept for CAPFs, Armed Forces and State Police.

These benefits are the intended impacts described by the submitted SIH material; they should not be interpreted as measured field outcomes from a deployed operational system.

13. Current Prototype vs. Proposed Production System

Aspect	Visible / demonstrated in prototype	Proposed or requires further implementation
User experience	Role-based web interface, check-in, dashboards and intervention dialogs.	Production-grade authentication, authorization and accessibility validation.
Risk analytics	Risk indicators, longitudinal trend UI and example diagnostic attribution.	Validated model training, calibration, evaluation metrics and monitoring.
Data	Demonstration records and simulated operational indicators are visible in the interface.	Secure integration with real HRMS/wellness/wearable sources.
AI stack	Website exposes AI-oriented functionality and the SIH document lists candidate model families.	Model-specific deployment and MLOps need validation against real datasets.
Security	Privacy/consent concepts and air-gapped framing are visible.	Formal threat modeling, penetration testing, key management and compliance assessment.
Deployment	Website advertises desktop, PWA and offline modes.	Operational rollout, device management and secure update mechanisms.

14. Future Development Roadmap

Phase	Focus
Phase 1	Stabilize role-based UI, data contracts, authentication and secure storage.
Phase 2	Integrate controlled HRMS/wellness data pipelines and formalize feature engineering.
Phase 3	Train and validate risk models using representative, ethically governed datasets; establish precision/recall, calibration and drift monitoring.
Phase 4	Add secure wearable integration where consented and technically appropriate.
Phase 5	Security assessment, offline deployment hardening, audit logging and operational pilot.
Phase 6	Pilot evaluation, user feedback, model recalibration and controlled scale-up.

15. Conclusion

RakshakAI presents a unified concept for predictive personnel welfare monitoring, combining daily self-assessment, longitudinal risk analytics, role-specific intervention workflows and privacy-focused command visibility. The supplied website demonstrates a substantial interactive prototype, including Personnel, Welfare Officer, Medical Officer and Commander experiences, risk indicators, intervention routing, resilience tools and offline/installable deployment options. The SIH submission provides the broader proposed architecture, AI approach and privacy model.

The next engineering priority is to clearly separate prototype demonstration data from production data, validate risk models on representative datasets, and harden authentication, security, privacy, auditability and deployment before any real-world operational use.

16. Source Notes

Primary SIH source: "Problem Statement ID – SIH26186, Problem Statement Title – AI-Based Predictive Personnel Stress and Welfare Monitoring System for Uniformed Forces," Team ID 188472, Team Name ISM Troopers. The document contains the idea submission, technical approach, feasibility/viability, impact/benefits and references.

Prototype source: RakshakAI public website at spal01-jpg.github.io/rakshak-ai/. The site was reviewed for the visible user roles, workflows, dashboards, risk indicators, intervention features and deployment descriptions.

Documentation note: This report distinguishes between capabilities visibly demonstrated by the prototype and technologies/architecture proposed in the SIH submission. Where the supplied material does not establish a production implementation, this report does not claim one.